

MEDICORE PHARMACY SALES ANALYTICS REPORT

Sales, Inventory and Product Performance Analysis

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Microsoft Excel // SQLite // Microsoft Power BI

Analysis Period: January 2020 – December 2025

1.0 Introduction

The pharmaceutical industry generates large volumes of operational data daily through medicine sales, inventory transactions, pricing activities, and customer purchases. However, the true value of this data lies not in its collection but in its ability to support informed business decision-making.

Pharmacies operate in highly competitive environments where maintaining adequate medicine availability, controlling inventory costs, and meeting customer demand are essential to sustained business performance. As transaction volumes increase across multiple countries, product categories, and customer segments, management requires timely and accurate insights to monitor sales performance, optimize inventory, and identify opportunities for growth.

This project was undertaken to transform raw pharmacy sales data into meaningful business intelligence capable of supporting strategic and operational decisions. Through the application of data analytics techniques, key performance indicators were measured and analysed to evaluate revenue performance, product demand, regional sales distribution, customer purchasing behaviour, stock availability, and medicine expiry status.

The analysis was performed using SQLite for data preparation and querying, Microsoft Excel for preliminary data validation, and Microsoft Power BI for dashboard development and visualization. By integrating these tools, the project provides a comprehensive view of pharmacy sales and inventory performance while highlighting areas where improved operational planning and data-driven decision-making can enhance business outcomes.

2.0 Executive Summary

This report presents a comprehensive analysis of the **MediCore Pharmacy Sales Dataset**, covering pharmacy sales and inventory activities between **January 2020 and December 2025**. The analysis was conducted using Microsoft Excel for initial data review, SQLite for data querying and transformation, and Microsoft Power BI for interactive dashboard development and visualization.

The dataset captures pharmacy operations across **19 countries**, incorporating information on medicine categories, customer age groups, regional sales, units sold, pricing, inventory levels, and medicine expiry status. During the analysis period, the pharmacy generated approximately **\$4.13 billion** in total revenue from over **79 million units** of medicines sold, reflecting strong commercial performance across multiple international markets.

The analysis revealed that revenue generation is distributed across several therapeutic categories, reducing dependence on a single product group and contributing to a more diversified revenue structure. However, certain medicines consistently outperformed others in both revenue generation and sales volume, indicating that a relatively small group of products serves as the primary drivers of financial performance.

Regional and country-level analysis further demonstrated variations in sales performance across international markets. While some countries contributed substantially to total revenue, others recorded comparatively lower performance, highlighting opportunities to strengthen market penetration and improve product availability in underperforming regions.

Inventory analysis showed that the majority of medicines maintained healthy stock levels and remained within acceptable expiry periods, suggesting generally effective inventory management practices. Nevertheless, the dashboard also identified products approaching low-stock thresholds and medicines with limited remaining shelf life. Although these represent a relatively small proportion of total inventory, they require continuous monitoring to prevent stock shortages, minimise product wastage, and maintain uninterrupted product availability.

Overall, the findings indicate that the pharmacy demonstrates strong revenue-generating capability supported by diversified product categories and broad geographical coverage. Future performance improvements are likely to be achieved not through expansion alone, but through more effective inventory planning, proactive stock replenishment, optimized product allocation, and targeted strategies aimed at strengthening sales in lower-performing markets. These opportunities position the organization to improve operational efficiency, protect profitability, and support sustainable long-term growth.

Project Overview

Item	Description
Organisation	MediCore Pharmacy
Industry	Pharmaceutical Retail
Dataset	Global Pharmacy Sales Dataset
Analysis Period	2020–2025
Records Analysed	19 rows and 175,360 columnns
Tools Used	Microsoft Excel, SQLite, Microsoft Power BI

3.0 Project Objectives

The primary objective of this project is to develop an interactive Business Intelligence dashboard that transforms raw pharmacy sales and inventory data into meaningful insights capable of supporting strategic and operational decision-making.

The specific objectives of the project are to:

1. Analyse overall sales performance across the six-year study period (2020–2025).
2. Evaluate revenue trends and identify changes in business performance over time.
3. Compare sales performance across countries and regions to identify high-performing and underperforming markets.
4. Assess revenue contribution across different medicine categories.
5. Identify the highest-performing medicines based on revenue generated and units sold.
6. Analyse customer purchasing behaviour across different age groups.
7. Monitor inventory levels and identify medicines approaching low-stock conditions.
8. Evaluate medicine expiry status to support proactive inventory management.
9. Develop interactive Power BI dashboards that enable users to monitor business performance using dynamic filters and visualisations.
10. Generate practical business insights and recommendations that support improved sales performance, inventory optimisation, and strategic decision-making.

4.0 Scope and Timeframe of Analysis

This analysis covers pharmacy sales and inventory activities recorded between **January 2020 and December 2025**. The study focuses on evaluating sales performance, customer purchasing patterns, product performance, and inventory conditions using the variables contained within the dataset.

The analysis includes the following business dimensions:

1. Revenue performance
2. Units sold
3. Product categories
4. Individual medicines

5. Customer age groups
6. Countries
7. Regions
8. Inventory levels
9. Medicine expiry status
10. COVID and Non-COVID sales comparison

The project does not incorporate external economic factors such as inflation, currency exchange rates, healthcare policy changes, competitor activities, or seasonal disease outbreaks beyond the COVID indicator included within the dataset. Consequently, all findings and recommendations are based solely on the information available in the MediCore Pharmacy Sales Dataset.

5.0 Data Source and Structure

The dataset used for this project is the **MediCore Pharmacy Sales Dataset (2020–2025)** obtained from Kaggle. It is a synthetic dataset designed to simulate real-world pharmacy operations and support business intelligence, sales analysis, inventory management, and forecasting.

The dataset consists of daily pharmacy sales records captured across multiple countries and regions. Each record represents a medicine transaction and contains information relating to sales performance, customer demographics, inventory levels, and product characteristics.

The dataset contains the following variables:

Variable	Description
Date	Transaction date
Year	Year of transaction
Month	Month of transaction
Day	Day of transaction
Region	Geographic region where the sale occurred

Country	Country in which the transaction was recorded
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Category	Medicine category
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Medicine	Name of the medicine sold
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Age Group	Customer age category
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Units Sold	Quantity of medicines sold
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Unit Price	Selling price per unit
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Stock Level	Available inventory quantity
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Expiry Days Remaining	Number of days before medicine expiry
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COVID Flag	Indicates whether the medicine is COVID-related
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To support business analysis, an additional field (**Revenue**) was derived by multiplying **Units Sold** by **Unit Price**. This calculated field served as the primary performance metric throughout the dashboard analysis.

6.0 Data Cleaning and Preparation

Before conducting the analysis, the dataset underwent a structured data cleaning and preparation process in **Microsoft Excel** to improve data quality, ensure consistency, and prepare the data for SQL querying and Power BI visualization. Once the dataset was cleaned and validated, it was imported into **SQLite** for data analysis and subsequently into **Microsoft Power BI** for dashboard development.

The data preparation process involved the following activities:

6.1 Data Import and Initial Assessment

The MediCore Pharmacy Sales Dataset was downloaded in CSV format and imported into Microsoft Excel for preliminary assessment. An initial review was conducted to understand the structure of the dataset, identify the available variables, and assess the overall quality of the data before analysis.

During this stage, the dataset was examined for missing values, duplicate records, inconsistent formatting, and incorrect data types that could affect the accuracy of subsequent analyses.

6.2 Data Cleaning in Microsoft Excel

Data cleaning was performed in Microsoft Excel to improve the quality and reliability of the dataset. The following tasks were completed:

1. Checked for duplicate records and confirmed that the dataset contained no duplicate transactions.
2. Verified that all mandatory fields were populated and confirmed that there were no significant missing values requiring treatment.
3. Reviewed the formatting of date fields to ensure consistency throughout the dataset.
4. Examined numerical variables such as **Units Sold**, **Unit Price**, **Stock Level**, and **Expiry Days Remaining** to confirm that they contained valid numerical values.
5. Standardised column names and verified that categorical variables, including **Region**, **Country**, **Category**, **Medicine**, and **Age Group**, were consistently recorded.

Completing these steps ensured that the dataset was accurate, consistent, and suitable for further analysis.

6.3 Data Preparation in SQLite

After cleaning the dataset in Microsoft Excel, the refined dataset was imported into **DB Browser for SQLite** for data querying and analysis.

To support financial analysis, a new **Revenue** field was created by multiplying **Units Sold** by **Unit Price**. Since the imported revenue values initially contained currency symbols and commas, SQL functions were used to remove these characters and convert the values into a numeric format suitable for aggregation and analysis.

SQLite queries were then used to validate calculations and generate business metrics that formed the basis of the Power BI dashboard.

6.4 SQL Analysis

SQL was used to extract key performance indicators and answer important business questions relating to MediCore Pharmacy's operations. Queries were developed to analyse:

1. Total Revenue
2. Revenue by Year
3. Revenue by Country

4. Revenue by Region
5. Revenue by Medicine Category
6. Revenue by Customer Age Group
7. Top-Selling Medicines
8. Average Unit Price by Category
9. Low-Stock Medicines
10. Medicines Nearing Expiry
11. Inventory Performance

The outputs from these queries were validated before being imported into Microsoft Power BI for visualization.

6.5 Dashboard Development

Following data preparation and SQL analysis, the cleaned dataset was imported into **Microsoft Power BI** for dashboard development.

Several DAX measures were created to calculate key performance indicators, including **Total Revenue**, **Total Units Sold**, **Average Revenue per Transaction**, and **Number of Countries**. Interactive visualizations were then developed to present sales performance, product analysis, customer demographics, and inventory management in a clear and intuitive manner.

The dashboard was designed with two interactive pages:

1. **Executive Overview**, which provides a high-level summary of MediCore Pharmacy's sales performance through KPI cards, revenue trends, and geographical analysis.
2. **Product & Inventory Analysis**, which provides detailed insights into medicine performance, customer age groups, stock levels, inventory health, and expiry status.

Interactive slicers were incorporated to allow users to filter the dashboard by year, region, category, and expiry status, enabling flexible exploration of the data from multiple business perspectives.

7.0 Dashboard Analysis and Interpretation

Following data preparation and modelling, an interactive dashboard was developed in Microsoft Power BI to provide a comprehensive overview of MediCore Pharmacy's sales performance, customer purchasing patterns, and inventory management. The dashboard consists of two pages: **Executive Overview** and **Product & Inventory Analysis**. These dashboards enable management to monitor key performance indicators (KPIs), identify business trends, evaluate product performance, and make informed decisions using interactive filters and visualizations.

7.1 Executive Overview Dashboard

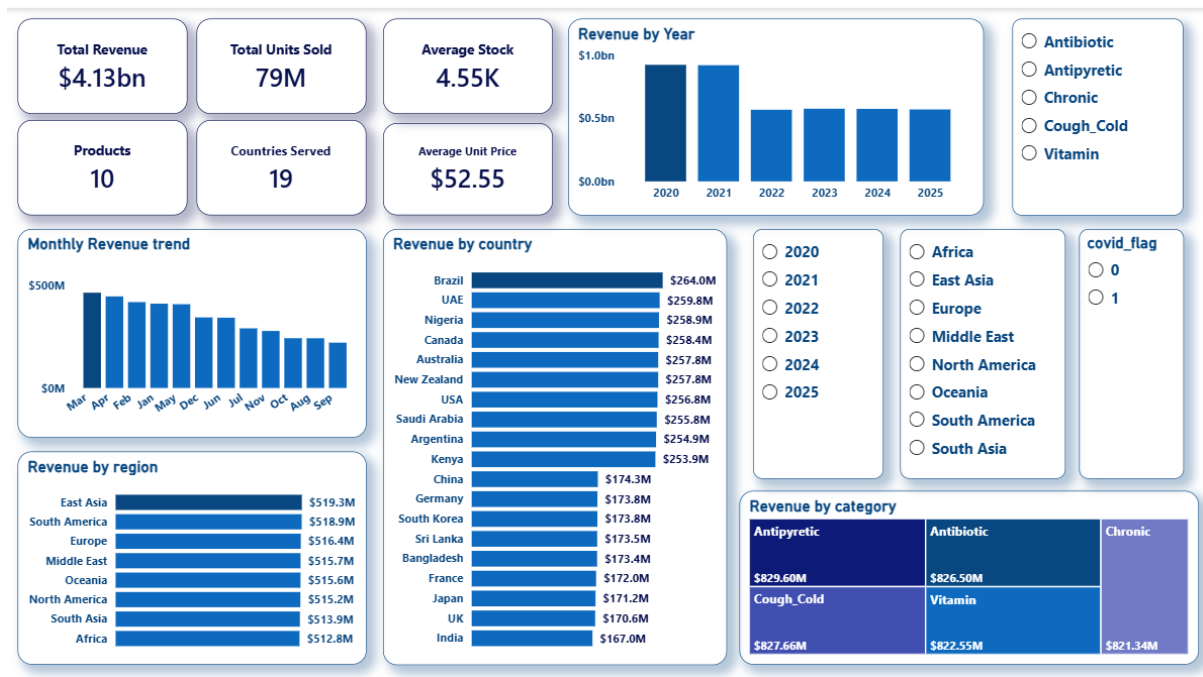


Figure 7.1: Executive Overview Dashboard

The Executive Overview dashboard provides a high-level summary of MediCore Pharmacy's overall business performance between 2020 and 2025. It presents key financial and operational indicators alongside interactive visualizations that allow management to evaluate sales trends, regional performance, and revenue distribution across product categories.

The dashboard includes KPI cards for Total Revenue, Total Units Sold, Number of Countries, and Average Revenue per Transaction. It also contains visualizations illustrating revenue trends over time, revenue by country, revenue by region, and revenue by medicine category. Interactive slicers allow users to analyse performance by year, region, and product category.

7.1.1 Overall Business Performance

The dashboard indicates that MediCore Pharmacy generated approximately **\$4.13 billion** in revenue from the sale of more than **79 million units** of medicines across **19 countries** during the six-year analysis period. The scale of these operations demonstrates that the organisation maintains a broad geographical presence while serving diverse customer needs across multiple therapeutic categories.

Insight

The combination of strong revenue, high sales volume, and international market coverage suggests that MediCore Pharmacy has established a diversified business model rather than relying on a single market or product line. This diversification reduces the organisation's exposure to risks associated with changes in demand within any one country or medicine category.

Recommendation

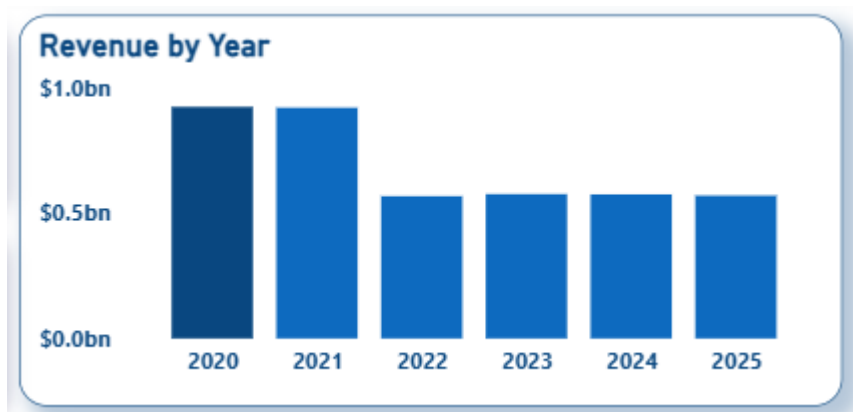
Management should continue strengthening its diversified operating model while identifying the products and markets responsible for the largest share of revenue. Investment decisions should prioritise areas that consistently demonstrate strong commercial performance without neglecting emerging markets that have the potential for future growth.

How to Achieve It

1. Conduct quarterly performance reviews using the Power BI dashboard to identify the highest-contributing countries and medicine categories.
2. Allocate procurement budgets based on historical demand rather than equal distribution across all products.
3. Develop country-specific sales targets that reflect each market's historical performance and growth potential.
4. Monitor KPI performance monthly to identify significant changes before they affect overall revenue.

7.1.2 Revenue Trend Analysis

The revenue trend chart presents MediCore Pharmacy's sales performance across the six-year study period. Revenue increased significantly during the earlier years before stabilising between 2022 and 2025.



Insight

The stabilisation of revenue suggests that while the pharmacy has maintained a strong market position, the business is no longer experiencing the rapid growth observed in the earlier years. Stable revenue is generally positive because it reflects consistent customer demand; however, prolonged stagnation may indicate that existing markets have reached maturity. Without new growth strategies, future revenue expansion may become increasingly difficult.

Recommendation

Management should focus on creating new revenue opportunities rather than relying solely on existing customer demand. This includes strengthening underperforming markets, introducing targeted promotional campaigns, and continuously reviewing product performance to identify additional growth opportunities.

How to Achieve It

1. Analyse historical revenue trends for each country to identify markets with declining or stagnant performance.
2. Introduce seasonal marketing campaigns for product categories that demonstrate lower sales.
3. Review pricing strategies periodically to ensure competitiveness without compromising profitability.
4. Compare year-on-year sales performance during quarterly management meetings to evaluate the effectiveness of growth initiatives.

7.1.3 Revenue by Country

The country analysis compares revenue generated across the nineteen countries in which MediCore Pharmacy operates. The visualization reveals noticeable differences in revenue contribution, indicating that sales performance varies significantly across markets. While several countries consistently generate substantial revenue, others contribute comparatively less despite operating within the same business model.



Insight

The disparity in country-level revenue suggests that market performance is influenced by factors beyond product availability alone. Countries such as **Brazil, the United Arab Emirates, Nigeria, Canada, and Germany** emerged as the strongest contributors to overall revenue,

indicating robust customer demand, effective product distribution, or stronger market penetration.

Conversely, the lower-performing countries contributed a considerably smaller share of total revenue. Although these markets continue to support the organisation's global presence, their relatively weak financial contribution indicates unrealised growth potential rather than poor business viability.

The variation between the highest and lowest revenue-generating countries presents an opportunity for MediCore Pharmacy to evaluate why successful markets consistently outperform others. Understanding these differences will enable management to replicate successful business practices across underperforming markets while preserving the strengths of existing high-performing locations.

Recommendation

Rather than implementing a uniform sales strategy across all countries, MediCore Pharmacy should adopt a differentiated market approach. High-performing countries should focus on maintaining customer loyalty and uninterrupted product availability, while lower-performing markets should receive targeted commercial interventions aimed at increasing demand and improving market penetration.

How to Achieve It

1. Conduct comparative analyses between the top-performing and lower-performing countries to identify differences in medicine demand, pricing, and purchasing behaviour.
2. Expand the availability of high-performing medicine categories in markets where customer demand remains underdeveloped.
3. Develop country-specific promotional campaigns that reflect local healthcare needs and purchasing patterns rather than applying identical marketing strategies across all regions.
4. Monitor country-level performance quarterly using the Power BI dashboard to evaluate the effectiveness of implemented strategies and adjust business plans where necessary.

Comparative Market Assessment

To support strategic decision-making, countries may be grouped according to their revenue performance rather than being evaluated individually.

**Performance
Category**

Strategic Interpretation

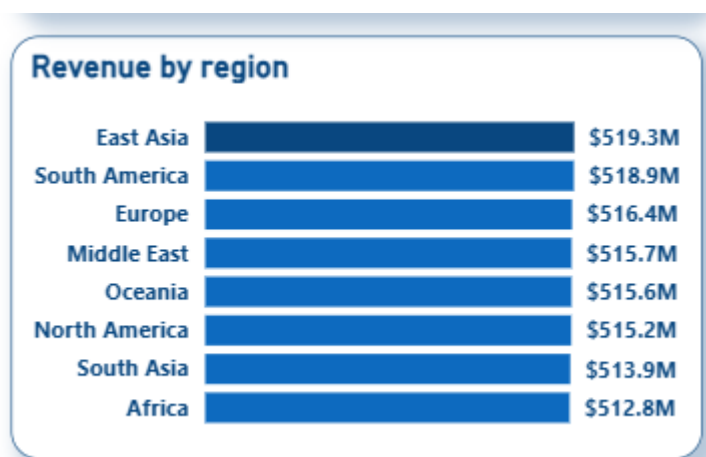
Management Priority

High Revenue Markets	Strong customer demand and mature market presence. These countries represent MediCore Pharmacy's primary revenue drivers.	Maintain product availability, strengthen customer retention, and protect market share.
Medium Revenue Markets	Stable performance with opportunities for further expansion.	Improve marketing activities, optimise inventory allocation, and identify products with growth potential.
Lower Revenue Markets	Comparatively lower contribution to overall revenue but potential for future growth.	Investigate operational constraints, review pricing strategies, strengthen product distribution, and increase market awareness.

This classification enables management to allocate resources according to business performance rather than distributing investment equally across all markets. Such an approach improves resource utilisation while supporting sustainable revenue growth across MediCore Pharmacy's international operations.

7.1.4 Revenue by Region

The regional analysis evaluates sales performance across the major geographical regions represented within the dataset. Revenue contributions are relatively balanced, although certain regions demonstrate stronger financial performance than others.



Insight

The relatively balanced regional distribution indicates that MediCore Pharmacy is not excessively dependent on a single geographical market. This diversified revenue structure

strengthens business resilience by reducing the financial impact of regional economic fluctuations or temporary changes in customer demand.

Recommendation

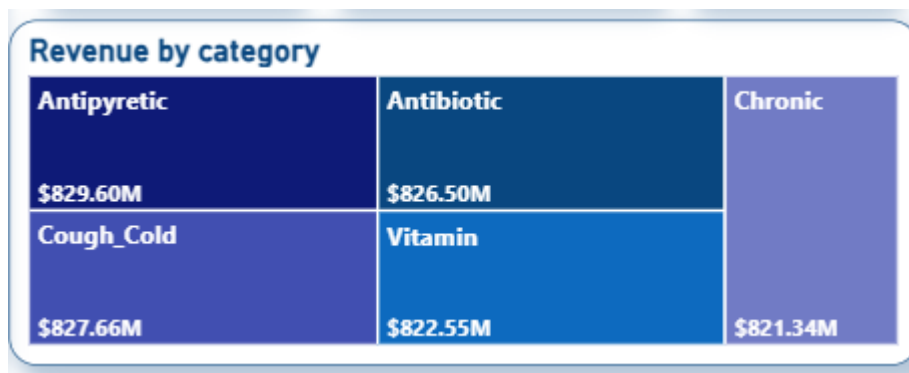
Management should preserve this balanced regional performance while directing additional resources toward regions with untapped growth potential.

How to Achieve It

1. Monitor regional sales performance monthly to identify emerging trends.
2. Allocate inventory based on regional demand rather than equal distribution.
3. Strengthen distribution networks in regions demonstrating increasing demand.
4. Conduct periodic regional performance reviews to identify operational challenges affecting sales.

7.1.5 Revenue by Medicine Category

The treemap illustrates the contribution of each medicine category to overall revenue. Revenue is distributed relatively evenly across the different categories, indicating that multiple product groups contribute meaningfully to business performance.



Insight

A balanced revenue contribution across medicine categories is a positive indicator of portfolio diversification. This reduces the organisation's dependence on any single therapeutic category and helps maintain financial stability even if demand for one category temporarily declines.

However, balanced revenue should not lead to equal investment across all categories. Some categories may generate similar revenue while requiring significantly different inventory levels, storage costs, or procurement investments. Understanding these operational differences is essential for maximising profitability.

Recommendation

Management should continue maintaining a diversified product portfolio while regularly evaluating the profitability and inventory efficiency of each medicine category. Investment

decisions should consider not only revenue contribution but also inventory turnover, stock holding costs, and customer demand.

How to Achieve It

- 1. Conduct monthly category performance reviews combining revenue, units sold, and inventory levels.
- 2. Prioritise procurement for categories demonstrating both high revenue and strong inventory turnover.
- 3. Reduce excess purchasing for categories experiencing slower inventory movement.
- 4. Monitor customer demand trends to ensure procurement decisions remain aligned with changing purchasing behaviour.

7.1.6 Executive Dashboard Summary

The Executive Overview dashboard demonstrates that MediCore Pharmacy maintains strong financial performance supported by diversified revenue streams, broad international operations, and balanced product category contributions. While the business exhibits stable revenue generation and healthy market coverage, the analysis also highlights opportunities to strengthen performance through market-specific strategies, improved inventory planning, and continuous monitoring of key performance indicators.

The insights generated from this dashboard provide management with a strategic overview of organisational performance and establish the foundation for the more detailed product and inventory analysis presented in the next section.

7.2 Product & Inventory Analysis Dashboard



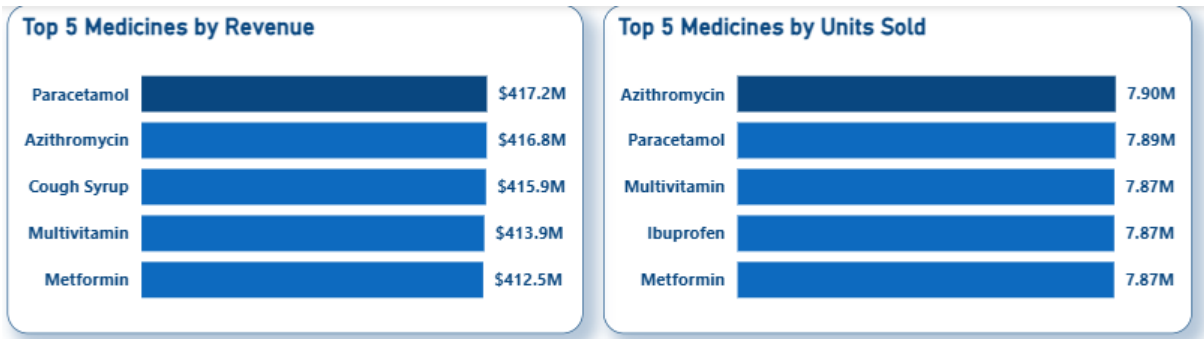
Figure 7.2: Product & Inventory Analysis Dashboard

The Product & Inventory Analysis dashboard provides a detailed evaluation of MediCore Pharmacy's product performance and inventory health. While the Executive Overview dashboard presents the organisation's overall performance, this dashboard focuses on identifying the medicines, customer segments, and inventory conditions that drive operational success.

The dashboard includes visualisations showing the top-performing medicines, revenue distribution by customer age group, stock levels across regions, inventory expiry status, and total stock availability. Together, these visualisations enable management to assess product demand, monitor inventory efficiency, and identify potential operational risks before they impact business performance.

7.2.1 Top Performing Medicines

The dashboard identifies the medicines that generated the highest revenue and sales volume throughout the analysis period. These products represent the strongest contributors to MediCore Pharmacy's overall financial performance and provide valuable insight into customer demand and purchasing behaviour.



Insight

The analysis reveals that a relatively small number of medicines account for a significant proportion of total revenue and units sold. Products such as **Paracetamol**, **Azithromycin**, and other top-performing medicines consistently outperform the remainder of the product portfolio, indicating sustained customer demand and high market acceptance.

While these products strengthen the pharmacy's financial performance, they also highlight a degree of **revenue concentration**. When a large share of revenue depends on a limited number of medicines, any disruption in their availability, such as supplier delays, regulatory changes, or sudden shifts in demand, can have a direct impact on overall business performance.

At the same time, the dashboard shows that several medicines contribute comparatively less to revenue despite occupying inventory space. This presents an opportunity to review the efficiency of inventory investment and determine whether these products continue to justify their current stock levels.

Business Implication

The current product portfolio demonstrates strong demand for essential medicines, which supports stable revenue generation. However, depending heavily on a small group of products increases operational and financial risk. Diversifying revenue across a broader range of medicines while protecting the availability of existing top performers will strengthen MediCore Pharmacy's long-term resilience.

Recommendation

MediCore Pharmacy should continue prioritising the availability of high-performing medicines while regularly evaluating lower-performing products to determine whether adjustments to procurement, pricing, or promotional strategies are required. Inventory investment should reflect both customer demand and product profitability rather than maintaining equal stock levels across all medicines.

How to Achieve It

1. Classify medicines into **High**, **Medium**, and **Low** revenue tiers using dashboard insights.
2. Establish automatic reorder points for high-performing medicines to minimise stock shortages.
3. Review slow-moving medicines every quarter to determine whether inventory levels should be reduced.
4. Analyse supplier performance to ensure uninterrupted availability of strategically important products.
5. Introduce targeted promotions for underperforming medicines where appropriate to improve inventory turnover before considering reductions in procurement.

Comparative Product Performance Assessment

To support procurement and inventory planning, medicines can be grouped according to their contribution to business performance.

Product Performance	Business Interpretation	Recommended Action
High-Performing Medicines	Generate a significant share of revenue and experience consistent customer demand.	Prioritise procurement, maintain safety stock, and closely monitor supplier performance.
Moderate-Performing Medicines	Contribute steadily to revenue but demonstrate moderate demand.	Maintain balanced inventory levels and monitor changes in customer purchasing behaviour.

Low-Performing Medicines	Generate relatively low revenue while occupying storage capacity.	Review demand trends, optimise procurement quantities, and consider promotional strategies to improve inventory turnover.
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This classification enables MediCore Pharmacy to allocate inventory investment more effectively while reducing unnecessary carrying costs and ensuring that high-demand medicines remain consistently available.

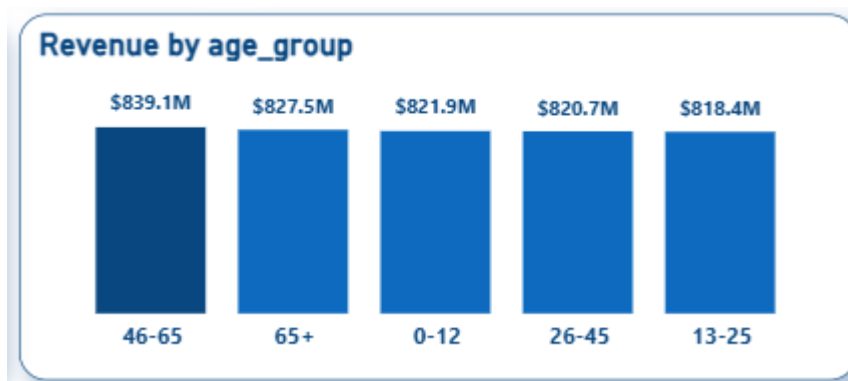
Strategic Observation

One of the most significant findings from this analysis is that **high sales volume does not always translate into maximum profitability**. Some medicines may sell frequently because they are low-cost products, while others generate greater revenue despite lower sales volumes due to higher unit prices.

Consequently, procurement decisions should not rely solely on units sold. Revenue, profit potential, inventory turnover, and customer demand should be evaluated together when determining purchasing priorities. This balanced approach enables MediCore Pharmacy to maximise financial performance while maintaining efficient inventory management.

7.2.2 Revenue by Customer Age Group

The dashboard analyses revenue generated across different customer age groups to understand purchasing behaviour and identify the demographic segments that contribute most to MediCore Pharmacy's overall sales performance.



Insight

The analysis indicates that customers aged **46–65 years** generated the highest proportion of revenue during the study period, followed closely by customers within the **26–45 years** age group. This trend suggests that middle-aged and older adults represent MediCore Pharmacy's primary customer base.

The higher purchasing activity within these age groups is likely associated with increased demand for medications used in the management of chronic conditions, preventive healthcare, and long-term treatments. Their consistent purchasing behaviour provides the pharmacy with a stable and predictable source of revenue.

In contrast, younger customers contribute a smaller proportion of total revenue. While this does not necessarily indicate poor performance, it suggests that this demographic may purchase medicines less frequently or primarily for short-term healthcare needs.

Business Implication

The concentration of revenue within a limited number of customer age groups demonstrates that MediCore Pharmacy benefits from a loyal and consistent customer base. However, excessive dependence on one demography increases business risk should purchasing behaviour change due to economic conditions, healthcare policies, or demographic shifts.

Broadening customer engagement across multiple age groups will create a more balanced and sustainable revenue structure.

Recommendation

MediCore Pharmacy should continue strengthening relationships with its highest-value customer segments while simultaneously expanding its engagement with younger customers through targeted healthcare initiatives and preventive wellness programmes.

How to Achieve It

1. Develop loyalty programmes that reward repeat purchases among existing customers.
2. Ensure consistent availability of medicines commonly purchased by customers aged 46–65 years.
3. Introduce wellness campaigns focusing on preventive healthcare for younger adults.
4. Analyse customer purchasing patterns quarterly to identify emerging demographic trends and adjust product offerings accordingly.
5. Partner with healthcare providers to promote routine health screenings that encourage earlier engagement with pharmacy services.

Customer Segment Performance Assessment

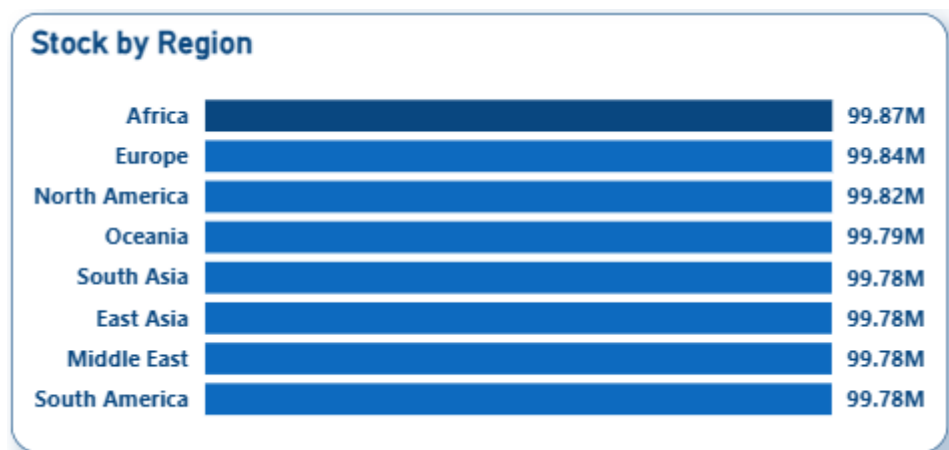
Customer Segment	Business Interpretation	Recommended Action
High Revenue Age Groups	Stable demand and consistent purchasing behaviour.	Maintain product availability, strengthen customer loyalty, and prioritise service quality.

Moderate Revenue Age Groups	Demonstrate steady purchasing activity with growth potential.	Expand targeted health education and promotional campaigns.
Lower Revenue Age Groups	Limited contribution to current revenue but represent future growth opportunities.	Increase awareness through preventive healthcare initiatives and digital engagement strategies.

This approach enables MediCore Pharmacy to strengthen relationships with its most valuable customers while creating opportunities to expand demand among underrepresented demographic groups.

7.2.3 Inventory Levels Across Regions

The inventory analysis compares stock availability across MediCore Pharmacy's operating regions to evaluate the effectiveness of inventory distribution and support efficient supply chain management.



Insight

The dashboard indicates that inventory is distributed relatively evenly across regions, reflecting a structured inventory allocation process. However, regional sales performance differs across markets, meaning that equal stock distribution does not necessarily translate into optimal inventory management.

Regions generating stronger sales are likely to experience faster inventory turnover than lower-performing regions. Maintaining identical stock quantities across all locations may therefore result in excess inventory in slower markets while increasing the likelihood of stock shortages in higher-demand areas.

Business Implication

Inventory should follow customer demand rather than geographical equality. Aligning stock allocation with actual purchasing behaviour will improve inventory turnover, reduce holding costs, and minimise the risk of stock shortages.

Recommendation

Inventory planning should become demand-driven rather than distribution-driven. Procurement and replenishment decisions should consider historical sales performance, regional demand patterns, and inventory turnover instead of allocating similar stock quantities across all regions.

How to Achieve It

1. Review regional inventory turnover on a monthly basis.
2. Increase replenishment frequency in regions demonstrating consistently high sales.
3. Redistribute excess inventory from lower-demand regions before placing additional purchase orders.
4. Integrate regional sales trends into procurement planning to improve forecasting accuracy.
5. Establish regional inventory performance targets and monitor them through Power BI dashboards.

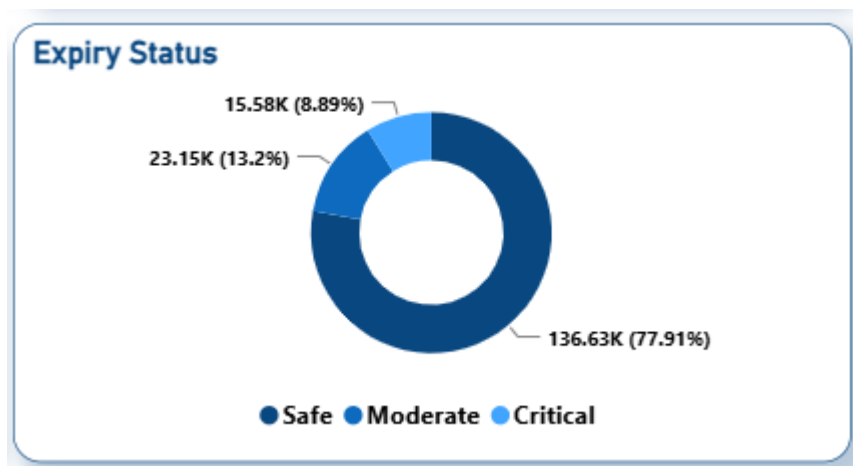
Regional Inventory Assessment

Inventory Status	Business Interpretation	Recommended Action
High Stock with High Sales	Inventory supports customer demand effectively.	Maintain existing replenishment strategy while monitoring turnover.
High Stock with Low Sales	Potential overstocking and increased holding costs.	Reduce procurement and consider transferring stock to higher-demand regions.
Low Stock with High Sales	Increased risk of stock shortages and lost revenue.	Increase reorder frequency and strengthen demand forecasting.

This inventory classification helps MediCore Pharmacy allocate stock more efficiently while reducing unnecessary inventory costs.

7.2.4 Inventory Expiry Status

The dashboard categorises medicines according to the number of days remaining before expiry, providing an overview of inventory health and highlighting products that require immediate operational attention.



Insight

The analysis shows that the majority of medicines remain within the **Safe** expiry category, indicating that MediCore Pharmacy generally maintains effective inventory rotation practices. This reflects disciplined procurement planning and stock management.

However, the dashboard also identifies medicines within the **Moderate** and **Critical** expiry categories. Although these products represent a smaller proportion of total inventory, they require immediate attention because delayed action may result in expired medicines, financial losses, and unnecessary inventory waste.

An important observation is that products approaching expiry still occupy valuable storage space while generating limited financial value if they are not sold before expiration.

Business Implication

Inventory expiry is not only an operational concern but also a financial performance issue. Every medicine that expires before being sold represents lost revenue, increased procurement costs, and reduced inventory efficiency.

Effective expiry management therefore contributes directly to profitability by reducing avoidable losses and improving inventory turnover.

Recommendation

MediCore Pharmacy should strengthen its expiry management process by introducing continuous monitoring of medicines approaching expiry and integrating expiry information into routine inventory decision-making.

How to Achieve It

1. Generate weekly dashboard reports highlighting medicines within the Moderate and Critical expiry categories.
2. Implement the **First-Expiry, First-Out (FEFO)** inventory management principle across all locations.
3. Transfer near-expiry medicines from slower-selling regions to branches with higher customer demand.
4. Introduce controlled promotional pricing for medicines approaching expiry where appropriate.
5. Include expiry monitoring as a standing agenda item during monthly inventory review meetings.

Expiry Risk Assessment

Expiry Category	Business Interpretation	Recommended Action
Safe	Inventory remains suitable for normal distribution and sale.	Continue routine inventory monitoring.
Moderate	Products require closer observation to prevent future losses.	Prioritise sales and monitor weekly.
Critical	High probability of product expiry and financial loss.	Immediate intervention through redistribution or promotional clearance.

This structured approach allows MediCore Pharmacy to minimise avoidable product wastage while maintaining healthy inventory turnover.

7.2.5 Product & Inventory Dashboard Summary

The Product & Inventory Analysis dashboard provides a comprehensive assessment of MediCore Pharmacy's product portfolio and inventory performance. The analysis demonstrates that the organisation's revenue is primarily supported by a group of consistently high-performing medicines and a loyal customer base concentrated within the 46–65 age group. Inventory management practices are generally effective, as reflected by balanced stock distribution and a high proportion of medicines within the safe expiry category.

Despite these strengths, the dashboard identifies several opportunities for operational improvement. Revenue concentration among a limited number of medicines increases business risk, regional inventory allocation can be more closely aligned with customer demand, and medicines approaching expiry require more proactive management to minimise financial losses.

By integrating these insights into procurement planning, inventory management, and customer engagement strategies, MediCore Pharmacy can improve operational efficiency, strengthen revenue sustainability, and support long-term business growth while maintaining high levels of product availability and customer satisfaction.

8.0 Key Findings

The analysis of MediCore Pharmacy's sales and inventory data between 2020 and 2025 revealed several important findings regarding the organisation's operational performance, product demand, customer purchasing behaviour, and inventory management practices.

8.1 Strong Revenue Performance

MediCore Pharmacy generated approximately **\$4.13 billion** in revenue during the analysis period, demonstrating strong commercial performance across its international operations. The high sales value, combined with substantial units sold, indicates that the organisation maintains a healthy market presence and consistent customer demand.

The revenue trend further shows that although growth stabilised in the later years of the analysis period, the business maintained relatively consistent financial performance without significant fluctuations. This reflects a stable operating environment supported by continuous customer demand for pharmaceutical products.

8.2 Diversified Revenue Structure

Revenue was generated across multiple medicine categories, countries, and regions rather than depending on a single market or product group. This diversified revenue structure strengthens the organisation's financial resilience by reducing exposure to changes in demand within any individual market.

The analysis also indicates that while some countries contribute more significantly to overall revenue than others, MediCore Pharmacy has successfully established a broad international presence capable of supporting long-term business sustainability.

8.3 Revenue Concentration Among Key Medicines

The dashboard analysis revealed that a relatively small number of medicines generated a substantial proportion of total revenue and units sold. These products represent the pharmacy's primary revenue drivers and therefore play a critical role in overall business performance.

Although this demonstrates strong customer demand for these medicines, it also highlights a level of revenue concentration. Any disruption affecting the availability of these products could have a measurable impact on overall sales performance.

8.4 Customer Purchasing Behaviour

The analysis showed that customers aged **46–65 years** generated the highest revenue during the study period, followed by those within the **26–45 years** age group. These findings indicate that middle-aged adults remain MediCore Pharmacy's most valuable customer segment.

While this customer base provides stable revenue, younger customer groups contributed comparatively less to total sales, suggesting opportunities for future market expansion through targeted healthcare awareness and customer engagement initiatives.

8.5 Regional and Country Performance

Revenue analysis across countries and regions demonstrated that business performance varies considerably between markets. Certain countries consistently generated stronger financial performance, while others contributed relatively smaller shares of total revenue.

These differences suggest that market conditions, customer demand, and purchasing behaviour vary across geographical locations. Consequently, a uniform business strategy may not produce optimal results across all operating markets.

8.6 Effective Inventory Management

Inventory analysis indicates that MediCore Pharmacy generally maintains healthy stock availability across its operations. Most medicines remain within acceptable inventory levels, allowing the organisation to satisfy customer demand without widespread stock shortages.

The balanced inventory distribution observed across regions further suggests that inventory planning processes are functioning effectively.

8.7 Opportunities for Improved Expiry Management

Although the majority of medicines remain within the Safe expiry category, the dashboard identified products approaching Moderate and Critical expiry levels. While these medicines represent only a small proportion of total inventory, they present avoidable financial risks if not managed proactively.

Continuous monitoring of expiry status would further reduce inventory losses while improving stock utilisation.

8.8 Value of Business Intelligence

The development of the Power BI dashboard significantly improved the accessibility and interpretation of pharmacy performance data. Instead of relying on manual spreadsheet analysis, management can now monitor sales performance, inventory conditions, customer behaviour, and product demand through interactive dashboards that support faster and more informed decision-making.

Overall, the findings demonstrate that MediCore Pharmacy possesses strong commercial performance supported by diversified revenue streams, healthy inventory management practices, and broad geographical operations. However, the analysis also identifies

opportunities to improve demand forecasting, inventory optimisation, market-specific strategies, and proactive stock management to strengthen long-term business performance.

9.0 Recommendations

Based on the findings obtained from the analysis, the following recommendations are proposed to improve MediCore Pharmacy's operational efficiency, inventory management, and long-term business performance.

9.1 Adopt Data-Driven Procurement Planning

Procurement decisions should be guided by historical sales trends and product demand rather than relying solely on routine replenishment schedules. Medicines that consistently generate high revenue and sales volumes should receive procurement priority to minimise stock shortages while preventing unnecessary overstocking of slower-moving products.

How to Achieve This

1. Review monthly sales trends before placing procurement orders.
2. Establish minimum and maximum stock levels for each medicine based on historical demand.
3. Update procurement plans quarterly using insights generated from the Power BI dashboard.
4. Integrate demand forecasting into inventory planning to improve purchasing accuracy.

9.2 Implement Market-Specific Sales Strategies

Since revenue performance varies across countries and regions, MediCore Pharmacy should avoid applying identical commercial strategies across all markets. Business decisions should reflect local demand patterns, purchasing behaviour, and product preferences.

How to Achieve This

1. Analyse country-level performance reports on a quarterly basis.
2. Identify the medicine categories driving revenue in high-performing markets and assess their potential in lower-performing regions.
3. Develop targeted promotional campaigns that address the specific healthcare needs of individual markets.
4. Monitor regional performance using dashboard filters to evaluate the effectiveness of implemented strategies.

9.3 Strengthen Inventory Monitoring

Although inventory management is generally effective, continuous monitoring remains essential to maintaining product availability while reducing holding costs and preventing stock shortages.

How to Achieve This

1. Review inventory levels weekly using the dashboard.
2. Configure automatic alerts for medicines approaching minimum stock levels.
3. Compare inventory levels with sales performance before approving additional purchases.
4. Conduct periodic inventory audits to verify stock accuracy across all regions.

9.4 Improve Expiry Management Practices

Medicines approaching expiry should receive greater operational attention to minimise avoidable financial losses and improve inventory utilisation.

How to Achieve This

1. Generate weekly reports identifying medicines within the Moderate and Critical expiry categories.
2. Implement the First-Expiry, First-Out (FEFO) inventory management approach.
3. Redistribute near-expiry medicines to higher-demand locations where feasible.
4. Introduce controlled promotional discounts for selected medicines approaching expiry to accelerate sales.

9.5 Expand Customer Engagement Across Age Groups

While customers aged 46–65 years currently generate the highest revenue, MediCore Pharmacy should broaden its customer base by encouraging greater engagement among younger age groups.

How to Achieve This

1. Introduce preventive healthcare campaigns targeted at younger adults.
2. Expand awareness programmes promoting routine health screening and wellness products.
3. Develop customer loyalty programmes that encourage repeat purchases across all age groups.
4. Monitor demographic purchasing trends to evaluate changes in customer behaviour over time.

9.6 Institutionalise Business Intelligence

The Power BI dashboard should become an integral component of MediCore Pharmacy's routine management process rather than serving solely as a reporting tool. Continuous use of Business Intelligence will enable management to identify operational issues earlier, respond more effectively to changing market conditions, and make evidence-based decisions.

How to Achieve This

1. Present dashboard reports during monthly management meetings.
2. Assign responsibility for updating dashboard data after each reporting period.
3. Monitor key performance indicators regularly rather than relying on annual performance reviews.

4. Encourage department managers to incorporate dashboard insights into procurement, sales, and inventory planning.

Implementing these recommendations will enable MediCore Pharmacy to strengthen operational efficiency, improve inventory utilisation, optimise procurement decisions, and sustain long-term business growth through the effective use of Business Intelligence and data-driven decision-making.

10.0 Conclusion

This project successfully applied Business Intelligence techniques to analyse MediCore Pharmacy's sales and inventory performance using the Global Pharmacy Sales Dataset covering the period from 2020 to 2025. Through the combined use of Microsoft Excel, SQLite, and Microsoft Power BI, raw transactional data was transformed into interactive dashboards that provide meaningful insights into revenue performance, product demand, customer purchasing behaviour, and inventory management.

The analysis revealed that MediCore Pharmacy maintains strong financial performance supported by diversified revenue streams across multiple countries, regions, and medicine categories. The organisation also benefits from generally effective inventory management practices, with healthy stock availability and a high proportion of medicines remaining within safe expiry periods. These strengths position the pharmacy to sustain operational efficiency while meeting customer demand across its international markets.

Despite these positive outcomes, the analysis also identified opportunities for improvement. Revenue is concentrated among a relatively small group of medicines, highlighting the importance of ensuring continuous product availability and strengthening supplier relationships. Variations in country-level performance suggest that market-specific sales strategies could improve revenue generation in lower-performing regions, while continuous monitoring of medicines approaching expiry would further reduce avoidable inventory losses.

The Power BI dashboards developed in this project provide MediCore Pharmacy with a practical decision-support system that enables management to monitor key performance indicators in real time, identify emerging business trends, and respond proactively to operational challenges. By integrating data-driven insights into procurement planning, inventory management, and strategic decision-making, the organisation can improve resource utilisation, strengthen customer satisfaction, and support sustainable long-term growth.

Overall, this project demonstrates how Business Intelligence can transform routine operational data into actionable business insights. The combination of structured data analysis and interactive visualisation equips MediCore Pharmacy with the analytical capability required to make informed decisions, optimise business performance, and remain competitive within the pharmaceutical industry.